

Gambler's Ruin Problem

A gambler is playing with his opponent. The gambler has initial capital \$i\$ while the total capital is \$N\$. After each gamble, the gambler either wins \$1\$ or loses \$1\$ with probabilities \$p\$ and \$q = 1 - p\$ respectively. The game ends when he wins or loses all the money. Let

P_i = probability that the gambler wins all the money when he has \$i\$.

Then, we have

$$P_i = p P_{i+1} + q P_{i-1}$$

After some algebraic steps, we have

$$P_i = \begin{cases} \frac{1 - \left(\frac{q}{p}\right)^i}{1 - \left(\frac{q}{p}\right)^N}, & \text{if } p \neq q \\ \frac{i}{N}, & \text{if } p = q \end{cases}$$

Becoming infinitely rich or getting ruined

Let the total capital be infinite. If $p > q$, from the above equation we have:

$$\lim_{N \rightarrow \infty} P_i = 1 - \left(\frac{q}{p}\right)^i > 0$$

If $p \leq q$, we have

$$\lim_{N \rightarrow \infty} P_i = 0$$

That is, if the gambler's opponent has infinite capital, the gambler has a positive probability of winning (becoming infinitely rich) when $p > q$. However, if $p \leq q$, the probability of winning is zero.

Exercise:

John starts with \$2 and $p = 0.6$. What is the probability that John wins if the opponent also has \$2?

Solution:

$$P_2 = \frac{1 - \left(\frac{0.4}{0.6}\right)^2}{1 - \left(\frac{0.4}{0.6}\right)^4} = 0.91$$

Exercise:

In the problem above, what is the probability that John will become infinitely rich if the opponent has infinite capital?

Solution:

$$\lim_{N \rightarrow \infty} P_2 = 1 - \left(\frac{q}{p}\right)^2 = 1 - \left(\frac{0.4}{0.6}\right)^2 = 0.56$$

Random Walk Model

A simple one-dimensional random walk is a probabilistic walk on a number line: it starts at zero and at each step moves $+1$ or -1 with equal probability.

Example:

In the gambler's ruin problem with $p = 0.5$, the gambler's amount of profit after each gamble is a simple one-dimensional random walk. If $p \neq 0.5$, we can call it an asymmetric random walk.

Hitting probabilities in a random walk

Let a random walk start at zero. Let a and b be two positive integers. Let

$$p(a) = P(\text{process hits level 'a' before hitting level ' - b'})$$

If we compare with the gambler's ruin problem, we have $a = N - i$ and $b = i$ so that $N = a + b$. Thus,

$$p(a) = \begin{cases} \frac{1 - \left(\frac{q}{p}\right)^b}{1 - \left(\frac{q}{p}\right)^{a+b}}, & \text{if } p \neq q \\ \frac{b}{a+b}, & \text{if } p = q \end{cases}$$

Exercise:

You bought a share of stock for \$10. Stock price moves \$1 each day as a simple random walk. The probability of an increase is 0.55.

- i) What is the probability that the stock price will reach \$16 before it reaches \$5?
- ii) What is the probability that you will get infinitely rich?

Solution:

i)

We want the probability that the stock price goes up by 6 before going down by 5. That is, $a = 6$ and $b = 5$.

$$p(6) = \frac{1 - \left(\frac{0.45}{0.55}\right)^5}{1 - \left(\frac{0.45}{0.55}\right)^{11}} = 0.71$$

ii)

We use $i = 10$ in the gambler's ruin problem:

$$\lim_{N \rightarrow \infty} P_{10} = 1 - \left(\frac{q}{p}\right)^{10} = 1 - \left(\frac{0.45}{0.55}\right)^{10} = 0.86$$

Counting Process:

Let $\{N(t), t \geq 0\}$ represent the number of times an event occurs in time $(0, t]$. The process $N(t)$ is called a counting process if it satisfies the following two conditions:

1. $N(t)$ is a non-negative integer.
2. If $s < t$, then $N(s) \leq N(t)$.

Example:

- i) Number of customers arriving at a post office in the time interval $(0, t]$.
- ii) Number of accidents at a highway bend in the time interval $(0, t]$.

Poisson Process:

A counting process $\{N(t), t \geq 0\}$ is a Poisson process with parameter λ if

1. Distribution of $X(t + s) - X(t)$ is independent of t . (stationary increment)

2. Increments over non-overlapping intervals are independent. That is,

$$X(t + 2s) - X(t + s) \text{ is independent of } X(t + s) - X(t).$$

3. $N(0) = 0$. Also, when time interval h is small:

$$P(N(h) = 1) = \lambda h + o(h)$$

$$P(N(h) = 0) = 1 - \lambda h + o(h)$$

$$P(N(h) = j) = o(h), j \geq 2.$$

Theorem:

If a process $N(t)$ follows the above conditions, then

$$P(N(t) = k) = \frac{e^{-\lambda t} (\lambda t)^k}{k!}, \quad k = 0, 1, 2, \dots$$

Proof:

Let $p_k(t) = P(N(t) = k)$

Then,

$$p_k(t + h) = p_k(t) p_0(h) + p_{k-1}(t) p_1(h) + o(h)$$

$$= (1 - \lambda h) p_k(t) + \lambda h p_{k-1}(t) + o(h)$$

$$\therefore \frac{p_k(t + h) - p_k(t)}{h} = -\lambda p_k(t) + \lambda p_{k-1}(t) + \frac{o(h)}{h}$$

Taking limit ($h \rightarrow 0$) on both sides:

$$p'_k(t) = -\lambda p_k(t) + \lambda p_{k-1}(t) \dots \dots (A)$$

Proceeding in a similar way:

$$p'_0(t) = -\lambda p_0(t)$$

$$\therefore \frac{p'_0(t)}{p_0(t)} = -\lambda$$

$$\therefore \log_e p_0(t) = -\lambda t + c$$

$$\therefore p_0(t) = e^{-\lambda t} \text{ (by using } p_0(0) = 1)$$

By using $p_k(0) = 0$ for $k \geq 1$ and by induction, the solution to (A) is:

$$p_k(t) = \frac{e^{-\lambda t}(\lambda t)^k}{k!} \text{ (proved)}$$

Poisson-Exponential relation

Let $N(t)$ be a Poisson process with parameter λ . Let $X_1, X_2, \dots$ be a sequence of random variables representing inter-event times. That is, X_1 is the waiting time for the first event, X_2 is the waiting time between the first event and the second event, and so on. Then $X_n, n = 1, 2, \dots$, follows exponential distribution with parameter λ .

Proof:

Let the $(n - 1)$ st event occurs at time t . then,

$$P(X_n > x)$$

$$= P(\text{no occurrence between } (t, t + x])$$

$$= P(\text{no occurrence between } (0, x]) \text{ (since the process is stationary)}$$

$$= P(N(x) = 0)$$

$$= e^{-\lambda x}$$

Therefore, $X_n \sim \text{exponential}(\lambda)$ with mean $\frac{1}{\lambda}$.

Exercise:

Customers arrive at a post office according to a Poisson process with $\lambda = 10/\text{hour}$. Determine

- the expected number of customers in an 8-hour day.
- the probability that nobody arrives from 1:00 to 1:12 p.m.
- the average waiting time between two successive arrivals.

Solution:

a) $E(N(8)) = 8 \lambda = 80.$

b) Length of time: $12/60 \text{ hour} = 0.2 \text{ hour}.$

$$P(N(0.2) = 0) = e^{-10 \times 0.2} = 0.135$$

c) Waiting time, $X \sim \text{exponential}(\lambda = 10).$

$$E(X) = \frac{1}{\lambda} = \frac{1}{10} \text{ hour} = 6 \text{ minutes}.$$

Pure Birth Process

A pure birth process $N(t)$ is more general than a Poisson process. In a Poisson process, parameter λ is constant. In a pure birth process, parameter λ_n depends on population size n .

Conditions:

When time interval h is small:

$$P(N(h) = 1) = \lambda_n h + o(h)$$

$$P(N(h) = 0) = 1 - \lambda_n h + o(h)$$

$$P(N(h) = j) = o(h), j \geq 2.$$

Differential Equations:

By generalizing the Poisson process, we have

$$p'_n(t) = -\lambda_n p_n(t) + \lambda_{n-1} p_{n-1}(t)$$

$$p'_0(t) = -\lambda_0 p_0(t)$$

Yule Process

It is a special case of pure birth process where $\lambda_n = n \lambda$. Let the initial population be one. That is, $p_1(0) = 1$ and $p_0(0) = 0$. The differential equation is given by:

$$p'_n(t) = -n\lambda p_n(t) + (n-1)\lambda p_{n-1}(t)$$

It can be shown that

$$p_n(t) = (1 - e^{-\lambda t})^{n-1} e^{-\lambda t} ; n = 1, 2, 3, \dots$$

which is a geometric distribution with parameter $p = e^{-\lambda t}$ and mean $= \frac{1}{e^{-\lambda t}} = e^{\lambda t}$.

If the initial population is i , it can be shown that

$$p_n(t) = \binom{n-1}{i-1} e^{-i\lambda t} (1 - e^{-\lambda t})^{n-i} ; n = i, i+1, i+2, \dots$$

which is a negative binomial distribution with parameters $r = i$ and $p = e^{-\lambda t}$. The mean is obtained as

$$E(N(t)) = i e^{\lambda t}.$$

Exercise:

A vendor sells fish kababs as follows. Every day he goes to a new place and gives fish kababs to two customers for free. From then on, each customer brings new customers with a rate of 1/hour.

- Determine the probability that after 3 hours the total number of customers will be 7.
- What is the average number of customers during a 2-hour period?

Solution:

(a)

This is a Yule process with $i = 2$, $\lambda = 1/\text{hour}$.

$$\begin{aligned} p_7(3) &= \binom{7-1}{2-1} e^{-2 \times 1 \times 3} (1 - e^{-1 \times 3})^{7-2} \\ &= 0.0115 \end{aligned}$$

(b)

$$E(N(2)) = 2 e^{1 \times 2} = 14.78$$